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## Optimization Technique

\* Optimization:-

- Optimization refers to the process of making a system or design as effective or functional as possible. In mathematical terms, it involves finding the best solution from a set of feasible solutions.
- Optimization is used in various fields such as engineering, economics, logistics, ML etc. The goal is usually to minimize or maximize a particular function (called the objective function) under a given set of constraints.
- The optimization techniques provided are to <sup>time</sup> fine-tune your code after finalizing design strategy, Algorithm data structures and implementation.

\* Components of a optimization problem

Generally an optimization problem has 3 components

i) Minimize  $f(x)$  w.r.t  $x$  subject to  $a \leq x \leq b$

ii) The objective function:

- The first component is an objective function  $f(x)$  which we are trying to either maximize or minimize.
- In general, we talk about minimization problems, this is simply because, if you have a maximization problem with  $f(x)$  we can convert it to a minimization problem with  $-f(x)$  without loss of generality we can look at minimization problem.

2) Decisional variable  $(x)$ :

which the second component is the decision variable we can choose to minimize the function so we write this as  $\min f(x)$



3) Constraints:-

The 3<sup>rd</sup> component is the constraint which basically constrains this  $x$  to some set

\* Types of optimization problems

- Constraint optimization problems in

In cases where the constraint is given that we and we have to have the solution satisfy this constraint we call them constraint optimization problem

- Unconstraint optimization problem

In cases where the constraint is missing we call them unconstraint optimization problems.

\* Applications

- Portfolio optimization in finance and data driven investment analysis, optimization helps allocate resource across assets to maximize return and minimize risk

- clustering and classification algorithms like k-means and SVM rely heavily on optimization to define clusters or decision boundaries.

- Recommendation system: Optimization techniques are used to reduce prediction errors in collaborative filtering and matrix factorization.

- Feature selection: optimization helps in identifying the most relevant features that improve model performance.

- Hyperparameter tuning:- Optimization methods are used to find the best combination of model parameter for improve accuracy

### \* Classical methods:-

- Classical optimization methods are a set of techniques that utilize differential calculus to find the optimal solution for problem involving continuous and differential functions. These methods are analytical in nature and rely on finding stationary points (where the derivative is 0) and analyzing the 2nd order derivative to determine whether these points are maxima, minima or saddle points.
  - 3 main types of problem can be handled by the classical optimization technique:
    - Single variable functions
    - Multi-variable function with no constraints
    - Multi-variable function with both equality and inequality constraints
- in problems with equality constraints the Lagrange multiplier method can be used. If the problem has inequality constraints the Kuhn Tucker conditions can be used to identify the optimum solution.



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Q1: Using the method of Lagranges multiply, solve the following NLP (non-linear programming problem)

Optimize  $Z = 6x_1^2 + 5x_2^2$ , subject to  $x_1 + 5x_2 = 7$   
 $x_1, x_2 > 0$

Sol<sup>n</sup>: we have the Lagrange func<sup>n</sup>

$$L(x_1, x_2, \lambda) = 6x_1^2 + 5x_2^2 - \lambda(x_1 + 5x_2 - 7) \quad \text{--- (1)}$$

obtain the partial derivative of

$$\frac{\partial L}{\partial x_1} = 0, \quad \frac{\partial L}{\partial x_2} = 0, \quad \frac{\partial L}{\partial \lambda} = 0$$

differentiate w.r.t  $x_1$

$$\begin{aligned} \frac{dL}{dx_1} &= 12x_1 + 5 \frac{dx_2^2}{dx_1} - \lambda(1 + 5 \frac{dx_2}{dx_1}) \\ &= 12x_1 - \lambda \end{aligned}$$

$$\begin{aligned} \frac{dL}{dx_2} &= 0 + 10x_2 - \lambda(5) \\ &= 10x_2 - 5\lambda \end{aligned}$$

$$\begin{aligned} 12x_1 - \lambda &= 0 & 12x_1 &= \lambda \\ 10x_2 - 5\lambda &= 0 & 10x_2 &= 5\lambda \end{aligned}$$

$$\frac{dL}{d\lambda} = x_1 + 5x_2 - 7$$

$$\begin{aligned} 12x_1 - \lambda &= 0 & \lambda &= 12x_1 \\ 10x_2 - 5\lambda &= 0 & 5\lambda &= 10x_2, \lambda = 2x_2 \\ x_1 + 5x_2 - 7 &= 0 & x_1 + 5x_2 - 7 &= 0 \end{aligned}$$



$$\lambda = 12n_1 = 2n_2$$

$$12n_1 = 2n_2$$

$$6n_1 = n_2$$

$$n_1 + 5n_2 - 7 = 0$$

$$n_1 + 5(6n_1) - 7 = 0$$

$$n_1 + 30n_1 - 7 = 0$$

$$31n_1 = 7$$

$$n_1 = \frac{7}{31}$$

$$6\left(\frac{7}{31}\right) = n_2$$

$$n_2 = \frac{42}{31}$$

$$\lambda = 2(n_2)$$

$$= 2\left(\frac{42}{31}\right)$$

$$= \frac{84}{31}$$

$$f(n_1, n_2) = 6n_1^2 + 5n_2^2$$

$$h(n_1, n_2) = n_1 + 5n_2 - 7$$

|              | 0                                 | 1                                                                                                         | 2                                                                                                         |
|--------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| $\Delta = 0$ | 0                                 | $\frac{\partial h}{\partial n_1}$                                                                         | $\frac{\partial h}{\partial n_2}$                                                                         |
| 1            | $\frac{\partial h}{\partial n_1}$ | $\frac{\partial^2 f}{\partial n_1^2} - \lambda \frac{\partial^2 h}{\partial n_1^2}$                       | $\frac{\partial^2 f}{\partial n_1 \partial n_2} - \lambda \frac{\partial^2 h}{\partial n_1 \partial n_2}$ |
| 2            | $\frac{\partial h}{\partial n_2}$ | $\frac{\partial^2 f}{\partial n_2 \partial n_1} - \lambda \frac{\partial^2 h}{\partial n_2 \partial n_1}$ | $\frac{\partial^2 f}{\partial n_2^2} - \lambda \frac{\partial^2 h}{\partial n_2^2}$                       |

$$\frac{\partial L}{\partial x_1} = \Delta = \begin{vmatrix} 0 & 1 & 5 \\ 1 & 12 & 0 \\ 5 & 0 & 10 \end{vmatrix}$$

$$= 0 - 1(10) + 5(-60) \\ = -10 - 300 \\ = -310$$

$\therefore \Delta$  is negative,  $x_0$  is minima  
 $\therefore$  hence  $x_1 = \frac{7}{31}$ ,  $x_2 = \frac{42}{31}$ ,  $z_{\min} = 9$ .

$$z = 6\left(\frac{7}{31}\right)^2 + 5\left(\frac{42}{31}\right)^2 \\ = 6(0.0509) + 5(1.8355) \\ = 9.4829 \quad / \quad \frac{294}{31}$$

Q.2. Using the method of Lagrange solve the following NLPP.

optimize:  $z = 4x_1 + 8x_2 - x_1^2 - x_2^2$   
 subject to:  $x_1 + x_2 = 4$   
 $x_1, x_2 > 0$

$$L(x_1, x_2, \lambda) = (4x_1 + 8x_2 - x_1^2 - x_2^2) - \lambda(x_1 + x_2 - 4)$$

$$\frac{dL}{dx_1} = 4 - 2x_1 - \lambda \quad \frac{dL}{d\lambda} = x_1 + x_2 - 4$$

$$\frac{dL}{dx_2} = 8 - 2x_2 - \lambda$$



$$\begin{aligned} 4 - 2x_1 - \lambda &= 0, \quad \lambda = 4 - 2x_1 \\ 8 - 2x_2 - \lambda &= 0, \quad \lambda = 8 - 2x_2 \\ x_1 + x_2 - 4 &= 0 \end{aligned}$$

$$\lambda = 4 - 2x_1 = 8 - 2x_2$$

$$4 - 2x_1 = 8 - 2x_2$$

$$2x_1 - 2x_2 = 4 - 8$$

$$2(x_1 - x_2) = -4$$

$$x_1 - x_2 = -2$$

$$x_2 = x_1 + 2$$

$$x_1 + x_2 = 4$$

$$1 + x_2 - 4 = 0$$

$$x_1 + x_1 + 2 = 4$$

$$x_2 = 3$$

$$2x_1 = 2$$

$$x_1 = 1$$

$$\lambda = 4 - 2(1)$$

$$\lambda = 2$$

$$\Delta \in \mathbb{R}^1 \quad f(x) = f(x_1, x_2) = 4x_1 + 8x_2 - x_1^2 - x_2^2$$

$$h(x_1, x_2) = x_1 + x_2 - 4$$

|              |                                   |                                                                                                           |                                                                                                           |
|--------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| $\Delta = 0$ | 0                                 | $x_1$                                                                                                     | $x_2$                                                                                                     |
|              | 0                                 | $\frac{\partial h}{\partial x_1}$                                                                         | $\frac{\partial h}{\partial x_2}$                                                                         |
| $x_1$        | $\frac{\partial h}{\partial x_1}$ | $\frac{\partial^2 f}{\partial x_1^2} - \lambda \frac{\partial^2 h}{\partial x_1^2}$                       | $\frac{\partial^2 f}{\partial x_1 \partial x_2} - \lambda \frac{\partial^2 h}{\partial x_1 \partial x_2}$ |
| $x_2$        | $\frac{\partial h}{\partial x_2}$ | $\frac{\partial^2 f}{\partial x_1 \partial x_2} - \lambda \frac{\partial^2 h}{\partial x_1 \partial x_2}$ | $\frac{\partial^2 f}{\partial x_2^2} - \lambda \frac{\partial^2 h}{\partial x_2^2}$                       |

$$\Delta = \begin{vmatrix} 0 & 1 & 1 \\ 1 & -2 & 0 \\ 1 & 0 & -2 \end{vmatrix}$$

$$= 0 - 1(-2) + 1(2) \\ = 2 + 2 \\ = 4$$

$\therefore \Delta$  is positive  $x_0$  is maxima  
 $\therefore$  hence  $x_1 = 1$ ,  $x_2 = 3$ ,  $z = 7$

$$z = 4x_1 + 8x_2 - x_1^2 - x_2^2 \\ = 4(1) + 8(3) - 1^2 - 3^2 \\ = 4 + 24 - 1 - 9 \\ = 18$$

- \* Non-linear programming problem with 3 variables & equality constraint  
 if  $\Delta_3$  &  $\Delta_4$  are both negative then  $x_0$  is minima  
 if  $\Delta_3$  is positive and  $\Delta_4$  is -ve then  $x_0$  is maxima

Q.1. Using the method of Lagranges solve the following  
 Optimize  $z = x_1^2 + x_2^2 + x_3^2 - 10x_1 - 6x_2 - 4x_3$   
 subject to:-  $x_1 + x_2 + x_3 = 7$

$$\frac{dL}{dx} = (x_1^2 + x_2^2 + x_3^2 - 10x_1 - 6x_2 - 4x_3) - \lambda(x_1 + x_2 + x_3 - 7)$$

$$\frac{dL}{dx_1} = 2x_1 - 10 - \lambda, \quad \frac{dL}{dx_2} = 2x_2 - 6 - \lambda$$

$$\frac{dL}{dx_3} = 2x_3 - 4 - \lambda, \quad \frac{dL}{d\lambda} = x_1 + x_2 + x_3 - 7$$



$$2x_1 - 10 - \lambda = 0$$

$$\lambda = 2x_1 - 10$$

$$2x_2 - 6 - \lambda = 0$$

$$\lambda = 2x_2 - 6$$

$$2x_3 - 4 - \lambda = 0$$

$$\lambda = 2x_3 - 4$$

$$x_1 + x_2 + x_3 = 7$$

$$2x_1 - 10 = 2x_2 - 6 = 2x_3 - 4 = 0$$

$$2x_1 - 2x_2 = 4$$

$$2x_2 - 6 = 2x_3 - 4$$

$$x_1 - x_2 = 2$$

$$2x_2 - 2x_3 = 2$$

$$x_1 = x_2 + 2$$

$$x_2 - x_3 = 1$$

$$x_2 = x_3 + 1$$

$$x_1 + x_2 + x_3 = 7$$

$$2x_1 - 10 = 2x_3 - 4$$

$$x_3 + 3 + x_3 + 1 + x_3 = 7$$

$$2x_1 - 2x_3 = 6$$

$$3x_3 + 4 = 7$$

$$x_1 - x_3 = 3$$

$$3x_3 = 3$$

$$x_1 = x_3 + 3$$

$$\boxed{x_3 = 1}$$

$$x_2 = x_3 + 1$$

$$x_1 = x_3 + 3$$

$$x_2 = 1 + 1$$

$$x_1 = 1 + 3$$

$$\boxed{x_2 = 2}$$

$$\boxed{x_1 = 4}$$

$$\lambda = 2x_1 - 10$$

$$= 2(4) - 10$$

$$= 8 - 10$$

$$\lambda = -2$$

$$f(x) = x_1^2 + x_2^2 + x_3^2 - 10x_1 - 6x_2 - 4x_3$$

$$h(x) = x_1 + x_2 + x_3 - 7$$

$$\Delta_4 = 0$$

|          | 0                                    | $\eta_1$                                                                                                              | $\eta_2$                                                                                                              | $\eta_3$                                                                                                              |
|----------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 0        | 0                                    | $\frac{\partial^2 h}{\partial \eta_1^2}$                                                                              | $\frac{\partial^2 h}{\partial \eta_1 \partial \eta_2}$                                                                | $\frac{\partial^2 h}{\partial \eta_1 \partial \eta_3}$                                                                |
| $\eta_1$ | $\frac{\partial h}{\partial \eta_1}$ | $\frac{\partial^2 f}{\partial \eta_1^2} - \lambda \frac{\partial^2 h}{\partial \eta_1^2}$                             | $\frac{\partial^2 f}{\partial \eta_1 \partial \eta_2} - \lambda \frac{\partial^2 h}{\partial \eta_1 \partial \eta_2}$ | $\frac{\partial^2 f}{\partial \eta_1 \partial \eta_3} - \lambda \frac{\partial^2 h}{\partial \eta_1 \partial \eta_3}$ |
| $\eta_2$ | $\frac{\partial h}{\partial \eta_2}$ | $\frac{\partial^2 f}{\partial \eta_1 \partial \eta_2} - \lambda \frac{\partial^2 h}{\partial \eta_1 \partial \eta_2}$ | $\frac{\partial^2 f}{\partial \eta_2^2} - \lambda \frac{\partial^2 h}{\partial \eta_2^2}$                             | $\frac{\partial^2 f}{\partial \eta_2 \partial \eta_3} - \lambda \frac{\partial^2 h}{\partial \eta_2 \partial \eta_3}$ |
| $\eta_3$ | $\frac{\partial h}{\partial \eta_3}$ | $\frac{\partial^2 f}{\partial \eta_1 \partial \eta_3} - \lambda \frac{\partial^2 h}{\partial \eta_1 \partial \eta_3}$ | $\frac{\partial^2 f}{\partial \eta_2 \partial \eta_3} - \lambda \frac{\partial^2 h}{\partial \eta_2 \partial \eta_3}$ | $\frac{\partial^2 f}{\partial \eta_3^2} - \lambda \frac{\partial^2 h}{\partial \eta_3^2}$                             |

$$\Delta_4 =$$

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 1 |
| 1 | 2 | 0 | 0 |
| 1 | 0 | 2 | 0 |
| 1 | 0 | 0 | 2 |

$$\Delta_3 =$$

|   |   |   |
|---|---|---|
| 0 | 1 | 1 |
| 1 | 2 | 0 |
| 1 | 0 | 2 |

$$= -4$$

$$= -1(2) + 1(-2) + 2(4)$$

$$= -2 - 2 + 8$$

$$= 4$$

$$\Delta_4 =$$

|   |    |    |    |
|---|----|----|----|
| 0 | 1  | 1  | -1 |
| 1 | 2  | 0  | 0  |
| 1 | -2 | 2  | 0  |
| 1 | 0  | -2 | 2  |

$$\Delta_4 =$$

|   |    |    |
|---|----|----|
| 1 | 2  | 0  |
| 1 | -2 | 2  |
| 1 | 0  | -2 |

$$= 1(4) - 2(-2) - 2(2)$$

$$= 4 + 4 - 4$$

$$= 4$$

$$= -12$$



Since both  $\Delta_4$  &  $\Delta_3$  are negative,  $X_0$  is minima  
 $\eta_1 = 4, \eta_2 = 2, \eta_3 = 1, \lambda = -2$

$$Z_{\min} = -35$$

Q. Determine the optimal sol<sup>n</sup> for the given problem & check whether it maximize or minimize the objective func<sup>n</sup>

Optimize;  $z = 2\eta_1^2 + 2\eta_2^2 + 2\eta_3^2 - 24\eta_1 - 8\eta_2 - 12\eta_3 + 260$

sub. to:  $\eta_1 + \eta_2 + \eta_3 = 11, \eta_1, \eta_2, \eta_3 > 0$

$$= (2\eta_1^2 + 2\eta_2^2 + 2\eta_3^2 - 24\eta_1 - 8\eta_2 - 12\eta_3 + 260) - \lambda(\eta_1 + \eta_2 + \eta_3 - 11)$$

$$\frac{dL}{d\eta_1} = 4\eta_1 - 24 - \lambda$$

$$\frac{dL}{d\eta_1} = 4\eta_1 - 24 - \lambda$$

$$\frac{dL}{d\eta_2} = 4\eta_2 - 8 - \lambda$$

$$\frac{dL}{d\eta_3} = 4\eta_3 - 12 - \lambda$$

$$\frac{dL}{d\lambda} = \eta_1 + \eta_2 + \eta_3 - 11$$

$$4\eta_1 - 24 - \lambda = 0$$

$$\lambda = 4\eta_1 - 24$$

$$4\eta_2 - 8 - \lambda = 0$$

$$\lambda = 4\eta_2 - 8$$

$$4\eta_3 - 12 - \lambda = 0$$

$$\lambda = 4\eta_3 - 12$$

$$\eta_1 + \eta_2 + \eta_3 = 11$$

$$4\eta_1 - 24 = 4\eta_2 - 8 = 4\eta_3 - 12$$

$$4\eta_2 - 8 = 4\eta_3 - 12$$

$$4\eta_1 - 24 = 4\eta_3 - 12$$

$$4\eta_2 - 4\eta_3 = -4$$

$$4\eta_1 - 4\eta_3 = +12$$

$$\eta_2 - \eta_3 = -1$$

$$\eta_1 - \eta_3 = +3$$

$$\eta_2 = \eta_3 - 1$$

$$\eta_1 = \eta_3 + 3$$

$$\eta_3 = 3$$

$$\eta_2 = \eta_3 - 1$$

$$\eta_2 = 2$$

$$= 3 - 1$$

$$\eta_1 = 6$$

$$\eta_1 + \eta_2 + \eta_3 = 11$$

$$+ \eta_3 + 3 + \eta_3 - 1 + \eta_3 = 11$$

$$3\eta_3 = 9$$

$$[\eta_2 = 2]$$

$$\lambda = 0$$

$$3\eta_3 + 2 = 11 \quad \eta_3 = \frac{9}{3} = 3$$

$$\eta_3 = 3$$

$$\begin{aligned}
 n_1 &= n_3 + 3 & n_1 &= n_3 + 3 \\
 &= 8 + 3 & &= 3 + 3 \\
 &= 11 & &= 6
 \end{aligned}$$

$$\begin{aligned}
 \lambda &= 4n_1 - 2n_2 \\
 &= 4(11) - 2n_2 \\
 &= 44 - 2n_2 = 0 \quad n_2 = 22
 \end{aligned}$$

$$\begin{aligned}
 f &= 2n_1^2 + 2n_2^2 + 2n_3^2 - 2n_1n_2 - 12n_1 - 12n_2 - 12n_3 \\
 &= 8n_3 + 260
 \end{aligned}$$

$$h = n_1 + n_2 + n_3 - 11$$

|              |                                   |                                                |                                                |                                                |
|--------------|-----------------------------------|------------------------------------------------|------------------------------------------------|------------------------------------------------|
| $\Delta h =$ | 0                                 | $\frac{\partial h}{\partial n_1}$              | $\frac{\partial h}{\partial n_2}$              | $\frac{\partial h}{\partial n_3}$              |
|              | $\frac{\partial h}{\partial n_1}$ | $\frac{\partial L}{\partial n_1^2}$            | $\frac{\partial L}{\partial n_1 \partial n_2}$ | $\frac{\partial L}{\partial n_1 \partial n_3}$ |
|              | $\frac{\partial h}{\partial n_2}$ | $\frac{\partial L}{\partial n_2 \partial n_1}$ | $\frac{\partial L}{\partial n_2^2}$            | $\frac{\partial L}{\partial n_3 \partial n_2}$ |
|              | $\frac{\partial h}{\partial n_3}$ | $\frac{\partial L}{\partial n_3 \partial n_1}$ | $\frac{\partial L}{\partial n_2 \partial n_3}$ | $\frac{\partial L}{\partial n_3^2}$            |

|              |   |   |   |   |
|--------------|---|---|---|---|
| $\Delta h =$ | 0 | 1 | 1 | 1 |
|              | 1 | 4 | 0 | 0 |
|              | 1 | 0 | 4 | 0 |
|              | 1 | 0 | 0 | 4 |
|              | + | - | + | - |

$$\begin{aligned}
 \Delta_3 &= \begin{vmatrix} 0 & 1 & 1 \\ 1 & 4 & 0 \\ 1 & 0 & 4 \end{vmatrix} \\
 &= -1(4) + 1(4) \\
 &= -4 - 4 \\
 &= -8
 \end{aligned}$$

$C_2 - C_3 \quad C_3 - C_4$

|              |   |    |    |   |
|--------------|---|----|----|---|
| $\Delta h =$ | 0 | 0  | 0  | 1 |
|              | 1 | 4  | 0  | 0 |
|              | 1 | -4 | 4  | 0 |
|              | 1 | 0  | -4 | 4 |

$$\begin{aligned}
 &= \begin{vmatrix} 1 & 4 & 0 \\ 1 & -4 & 4 \\ 1 & 0 & -4 \end{vmatrix} \\
 &= 1(16) - 4(-4 - 4) = 16 + 32 \\
 &= -16 + 32 \\
 &= 16 - 48 = -32
 \end{aligned}$$



Since  $\Delta_4, \Delta_3$  are -ve then it is minima

$$x_1 = 6, x_2 = 2, x_3 = 3, \lambda = 0$$

$$\begin{aligned} Z &= 2x_1^2 + 2x_2^2 + 2x_3^2 - 24x_1 - 12x_2 - 8x_3 + 260 \\ &= 2(6)^2 + 2(2)^2 + 2(3)^2 - 24(6) - 12(2) - 8(3) + 260 \\ &= 168 \end{aligned}$$

Q. Using the lagrange multiplier solve the given non-linear programming problem

maximize:-  $Z = 6x_1 + 8x_2 - x_1^2 - x_2^2$

Sub.to  $4x_1 + 3x_2 = 16, 3x_1 + 5x_2 = 15, x_1, x_2 > 0$

Soln:-  $L = 6x_1 + 8x_2 - x_1^2 - x_2^2 - \lambda_1(4x_1 + 3x_2 - 16) - \lambda_2(3x_1 + 5x_2 - 15)$

$$\frac{dL}{dx_1} = 6 - 2x_1 - 4\lambda_1 - 3\lambda_2 \quad \frac{dL}{dx_2} = 8 - 2x_2 - 3\lambda_1 - 5\lambda_2$$

$$\frac{dL}{d\lambda_1} = 4x_1 + 3x_2 - 16 \quad \frac{dL}{d\lambda_2} = 3x_1 + 5x_2 - 15$$

$$6 - 2x_1 - 4\lambda_1 - 3\lambda_2 = 0 \quad 2x_1 + 4\lambda_1 + 3\lambda_2 = 6 \quad \times 4$$

$$8 - 2x_2 - 3\lambda_1 - 5\lambda_2 = 0 \quad 2x_2 + 3\lambda_1 + 5\lambda_2 = 8 \quad \times 3$$

$$4x_1 + 3x_2 = 16$$

$$3x_1 + 5x_2 = 15$$

$$12x_1 + 9x_2 = 144$$

$$-12x_1 + 20x_2 = -60$$

$$-11x_2 = 84$$

$$x_2 = -\frac{84}{11}$$

$$3x_1 + 5\left(-\frac{84}{11}\right) = 15$$

$$3x_1 = 15 + \frac{420}{11}$$

$$= \frac{165 + 420}{11}$$

$$\begin{array}{r} 16 \\ \times 9 \\ \hline 144 \\ 144 \\ \hline 216 \\ \times 2 \\ \hline 432 \\ \hline 144 \\ \hline 60 \end{array}$$

$$\eta_1 = \frac{585}{11 \times 3}$$

$$= \frac{195}{11}$$

$$24 - 2(4\eta_1 + 3\eta_2) - 16\lambda_1 - 12\lambda_2$$

$$24 - 2(3\eta_1 + 5\eta_2) - 9\lambda_1 - 15\lambda_2$$

$$24 - 8\eta_1 - 16\lambda_1 - 12\lambda_2 + 48 - 6\eta_2 - 9\lambda_1 - 15\lambda_2 = 0$$

$$48 - 8\eta_1 - 6\eta_2 - 25\lambda_1 - 27\lambda_2 = 0$$

$$48 - 2(4\eta_1 + 3\eta_2) - 25\lambda_1 - 27\lambda_2 = 0$$

$$48 - 2(16) - 25\lambda_1 - 27\lambda_2 = 0$$

$$48 - 32 - 25\lambda_1 - 27\lambda_2 = 0$$

$$25\lambda_1 + 27\lambda_2 = 16 \quad \text{--- (1)}$$

$$6 - 2\eta_1 - 4\lambda_1 - 3\lambda_2 = 0 \quad \times 3$$

$$8 - 2\eta_2 - 3\lambda_1 - 5\lambda_2 = 0 \quad \times 5$$

$$18 - 6\eta_1 - 12\lambda_1 - 9\lambda_2 + 40 - 10\eta_2 - 15\lambda_1 - 25\lambda_2 = 0$$

$$58 - 8\eta_1 - 10\eta_2 - 27\lambda_1 - 34\lambda_2 = 0$$

$$58 - 2(3\eta_1 + 5\eta_2) - 27\lambda_1 - 34\lambda_2 = 0$$

$$58 - 2(15) - 27\lambda_1 - 34\lambda_2 = 0$$

$$58 - 30 - 27\lambda_1 - 34\lambda_2 = 0$$

$$27\lambda_1 + 34\lambda_2 = 28 \quad \text{--- (2)}$$

$$25\lambda_1 + 27\lambda_2 = 16 \quad \times 27$$

$$27\lambda_1 + 34\lambda_2 = 28 \quad \times 25$$

$$54\lambda_1 + 729\lambda_2 = 432$$

$$54\lambda_1 + 68\lambda_2 = 56$$



$$\begin{aligned} 675\lambda_1 + 729\lambda_2 &= 432 \\ -675\lambda_1 + 850\lambda_2 &= 700 \end{aligned}$$

$$-121\lambda_2 = 268$$

$$\lambda_2 = \frac{-268}{-121} = \frac{268}{121}$$

$$25\lambda_1 + 27\left(\frac{268}{121}\right) = 16$$

$$25\lambda_1 = 16 - \frac{7236}{121}$$

$$25\lambda_1 = \frac{1936 - 7236}{121}$$

$$\lambda_1 = \frac{-5300}{121 \times 25} = \frac{-212}{121}$$

$$6 - 2\eta_1 - 4\left(\frac{-212}{121}\right) - 3\left(\frac{268}{121}\right) = 0$$

$$2\eta_1 = 6 - 4\left(\frac{-212}{121}\right) - 3\left(\frac{268}{121}\right)$$

$$2\eta_1 = \frac{70}{11}$$

$$\eta_1 = \frac{35}{11}$$

$$8 - 2\eta_2 - 3\left(\frac{-212}{121}\right) - 5\left(\frac{268}{121}\right) = 0$$

$$2\eta_2 = 8 - 3\left(\frac{-212}{121}\right) - 5\left(\frac{268}{121}\right)$$

$$2\eta_2 = \frac{24}{11}$$

$$\eta_2 = \frac{12}{11}$$

Since  $n = 2$ ,  $m = 2$

$n - m = 0$ , the above method fails

find the signs of 2 principal minors of

$$A_{11} = \begin{vmatrix} \frac{\partial^2 Z}{\partial x_1^2} & \frac{\partial^2 Z}{\partial x_1 \partial x_2} \\ \frac{\partial^2 Z}{\partial x_1 \partial x_2} & \frac{\partial^2 Z}{\partial x_2^2} \end{vmatrix}$$

$$= \begin{vmatrix} -2 & 0 \\ 0 & -2 \end{vmatrix}$$

$$= 4$$

$$A_{22} = |a_{11}| = -2$$

$\therefore$  since  $a_{22}$  is -ve and  $a_{11}$  is +ve is maxima

$$x_1 = \frac{35}{11}$$

$$x_2 = \frac{12}{11}$$

$$\begin{aligned} Z &= 6x_1 + 8x_2 - x_1^2 - x_2^2 \\ &= 6\left(\frac{35}{11}\right) + 8\left(\frac{12}{11}\right) - \frac{35^2}{11} - \frac{12^2}{11} \\ &= \frac{1997}{121} \end{aligned}$$

Q. Using the method of Lagrange's solve

$$\text{optimize } Z = 4x_1^2 + 2x_2^2 + x_3^2 - 4x_1x_2$$

$$\text{Sub. to. } x_1 + x_2 + x_3 = 15$$

$$2x_1 - x_2 + 2x_3 = 20$$

$$x_1, x_2, x_3 > 0$$



$$L = 4x_1^2 + 2x_2^2 + x_3^2 - 4x_1x_2 - \lambda_1(x_1 + x_2 + x_3 - 15) - \lambda_2(2x_1 - x_2 + 2x_3 - 20)$$

$$\frac{dL}{dx_1} = 8x_1 - 4x_2 - \lambda_1 - 2\lambda_2 \quad \frac{dL}{dx_2} = 4x_2 - 4x_1 - \lambda_1 + \lambda_2$$

$$\frac{dL}{dx_3} = 2x_3 - \lambda_1 - 2\lambda_2 \quad \frac{dL}{d\lambda_1} = x_1 + x_2 + x_3 - 15$$

$$\frac{dL}{d\lambda_2} = 2x_1 - x_2 + 2x_3 - 20$$

$$8x_1 - 4x_2 - \lambda_1 - 2\lambda_2 = 0 \quad -1$$

$$4x_2 - 4x_1 - \lambda_1 + \lambda_2 = 0 \quad -2$$

$$2x_3 - \lambda_1 - 2\lambda_2 = 0$$

$$x_1 + x_2 + x_3 = 15$$

$$2x_1 - x_2 + 2x_3 = 20 \quad | \quad (2x_1 - x_2) = 20 - 2x_3$$

$$8x_1 - 4x_2 - \lambda_1 - 2\lambda_2 + 4x_2 - 4x_1 - \lambda_1 + \lambda_2 = 0$$

$$4x_1 - 2\lambda_1 - \lambda_2 = 0$$

$$2\lambda_1 + \lambda_2 = 4x_1$$

$$\lambda_1 + 2\lambda_2 = 2x_3$$

$$4 \left( 2x_1 - x_2 - \frac{\lambda_1}{2} - \frac{\lambda_2}{2} \right) = 0$$

$$4 \left( 20 - 2x_3 - \frac{\lambda_1}{2} - \frac{\lambda_2}{2} \right) = 0$$

$$80 - 8x_3 - \lambda_1 - 2\lambda_2 - 2x_3 + \lambda_1 + 2\lambda_2 = 0$$

$$80 - 10x_3 = 0$$

$$x_3 = \frac{80}{10} = x_3 = 8$$

$$\lambda_1 + \lambda_2 + 8 = 15$$

$$\lambda_1 + \lambda_2 = 7$$

$$2\lambda_1 - \lambda_2 + 2(8) = 20$$

$$2\lambda_1 - \lambda_2 = 4$$

$$3\lambda_1 = 11$$

$$\lambda_1 = \frac{11}{3}$$

$$\lambda_1 + \lambda_2 = 7$$

$$\frac{11}{3} + \lambda_2 = 7$$

$$\lambda_2 = 7 - \frac{11}{3}$$

$$\lambda_2 = \frac{21 - 11}{3} = \frac{10}{3}$$

$$8\left(\frac{11}{3}\right) - 4\left(\frac{10}{3}\right) - \lambda_1 - 2\lambda_2 = 0$$

$$\lambda_1 + 2\lambda_2 = 16 \quad \text{--- (1)}$$

$$2(8) = \lambda_1 + 2\lambda_2$$

$$\lambda_1 + 2\lambda_2 = 16 \quad \text{--- (2)}$$

$$4\left(\frac{10}{3}\right) - 4\left(\frac{11}{3}\right) - \lambda_1 + \lambda_2 = 0$$

$$\lambda_1 - \lambda_2 = -\frac{4}{3}$$

$$3\lambda_1 - 3\lambda_2 = -4 \quad \text{--- (2)}$$

$$3\lambda_1 + 6\lambda_2 = 48$$

$$+ 6\lambda_1 - 6\lambda_2 = -8$$

$$9\lambda_1 = 40$$

$$\lambda_1 = \frac{40}{9}$$

$$8\left(\frac{40}{9}\right) + 2\lambda_2 = 16$$

$$2\lambda_2 = 16 - \frac{40}{9}$$

$$\lambda_2 = \frac{104}{9 \times 2}$$

$$\begin{array}{r} 16 \\ 13 \\ \hline 48 \end{array}$$



$$\lambda_2 = \frac{52}{9}$$

(Since  $n = 3$ ,  $m = 2$

$n - m = 1$  the above method fails. Success

$$\text{Now, } P = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \frac{\partial h_1}{\partial x_2} & \frac{\partial h_1}{\partial x_3} \\ \frac{\partial h_2}{\partial x_1} & \frac{\partial h_2}{\partial x_2} & \frac{\partial h_2}{\partial x_3} \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 2 \end{bmatrix} \quad P^T = \begin{bmatrix} 1 & 2 \\ 1 & -1 \\ -1 & 2 \end{bmatrix}_{3 \times 2}$$

$$Q = \begin{bmatrix} \frac{\partial^2 L}{\partial x_1^2} & \frac{\partial^2 L}{\partial x_1 \partial x_2} & \frac{\partial^2 L}{\partial x_1 \partial x_3} \\ \frac{\partial^2 L}{\partial x_1 \partial x_2} & \frac{\partial^2 L}{\partial x_2^2} & \frac{\partial^2 L}{\partial x_2 \partial x_3} \\ \frac{\partial^2 L}{\partial x_1 \partial x_3} & \frac{\partial^2 L}{\partial x_2 \partial x_3} & \frac{\partial^2 L}{\partial x_3^2} \end{bmatrix}$$

$$= \begin{bmatrix} 8 & -4 & 0 \\ -4 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$$

$$H_2 = \begin{bmatrix} 0 & 0 & P^T \\ 0 & 0 & P^T \\ P^T & 0 \end{bmatrix}$$

$$H_6 = \begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & -1 & 2 \\ 1 & 2 & 8 & -4 & 0 \\ 1 & -1 & -4 & 4 & 0 \\ 1 & 2 & 0 & 0 & 0 \end{bmatrix}$$

$$(z - C_5, C_4 - C_5)$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -3 & 2 \\ 1 & 2 & 8 & -4 & 0 \\ 1 & -1 & -4 & 4 & 0 \\ 1 & 2 & 0 & 0 & 0 \end{bmatrix}$$

+   -   +   -   +

$$= +1 \begin{bmatrix} 0 & 0 & 0 & -3 \\ 1 & 2 & 8 & -4 \\ 1 & -1 & -4 & 4 \\ 1 & 2 & 0 & 0 \end{bmatrix}$$

+   -   +   -

$$= +3 \begin{bmatrix} 1 & 2 & 8 \\ 1 & -1 & -4 \\ 1 & 2 & 0 \end{bmatrix}$$

$$\begin{aligned} &= +3 [1(8) - 2(4) + 8(3)] \\ &= +3 [8 - 8 + 24] \\ &= +3 [24] \\ &= +72 \end{aligned}$$

\* Kuhn Tucker

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Q. Solve the following NLPP

$$\text{maximize } z: 10x_1 + 4x_2 + 2x_1^2 - x_2^2$$

$$\text{Subj to :- } 2x_1 + x_2 \leq 5 \quad x_1, x_2 \geq 0$$



soln:  $f(x_1, x_2) = 10x_1 + 4x_2 - 2x_1^2 - x_2^2$

$h(x_1, x_2) = 2x_1 + x_2 - 5$

- condn for Kuhn-Tucker

$$\frac{\partial f}{\partial x_1} - \lambda \frac{\partial h}{\partial x_1} = 0$$

$$\frac{\partial f}{\partial x_2} - \lambda \frac{\partial h}{\partial x_2} = 0$$

$$10 - 4x_1 - \lambda(2) = 0$$

$$4 - 2x_2 - \lambda(1) = 0$$

$$10 - 4x_1 - 2\lambda = 0 \quad \text{--- 1}$$

$$4 - 2x_2 - \lambda = 0 \quad \text{--- 2}$$

$$\lambda(2x_1 + x_2 - 5) = 0 \quad \text{--- 3}$$

$$2x_1 + x_2 - 5 \leq 0 \quad \text{--- 4}$$

$$\lambda, x_1, x_2 \geq 0 \quad \text{--- 5}$$

Case 1:  $\lambda = 0$

$$10 - 4x_1 = 0$$

$$4 - 2x_2 = 0$$

$$4x_1 = 10$$

$$2x_2 = 4$$

$$2x_1 = 5$$

$$x_2 = 2$$

$$x_1 = \frac{5}{2}$$

Put  $x_1, x_2$  in eqn 4

$$= 2(x_1) + x_2 - 5$$

$$= 2\left(\frac{5}{2}\right) + 2 - 5$$

$$= 7 - 5$$

$$= 2 \neq 0$$

This value do not satisfy eq<sup>n</sup> 4  
Hence  $\lambda = 0$  does not yield a feasible sol<sup>n</sup>  
so we reject this values

Case 2:  $2x_1 + x_2 - 5 = 0$

$$2x_1 + x_2 = 5$$

$$10 - 4x_1 - 2\lambda = 0$$

$$4 - 2x_2 - \lambda = 0$$

$$10 - 4x_1 - 2\lambda + 4 - 2x_2 - \lambda = 0$$

$$10 - 2(2x_1 + x_2) - 3\lambda + 4 = 0$$

$$10 - 2(5) - 3\lambda + 4 = 0$$

$$-3\lambda + 4 = 0$$

$$3\lambda = 4$$

$$\lambda = \frac{4}{3}$$

$$10 - 4x_1 - 2\lambda = 0$$

$$10 - 4x_1 - 2\left(\frac{4}{3}\right) = 0$$

$$4x_1 = 10 - \frac{8}{3}$$

$$4x_1 = \frac{22}{3}$$

$$x_1 = \frac{11}{6}$$

$$4 - 2(x_2) - \frac{4}{3} = 0$$

$$4 - 2x_2 - \frac{4}{3} = 0$$

$$2x_2 = 4 - \frac{4}{3}$$

$$2x_2 = \frac{8}{3}$$

$$x_2 = \frac{4}{3}$$



These values satisfy all the necessary conditions  
 $\therefore$  The optimal solution is  $x_1 = \frac{11}{6}$   $x_2 = \frac{4}{3}$

$$\begin{aligned} Z &= 10x_1 + 4x_2 - 2x_1^2 - x_2^2 \\ &= 10\left(\frac{11}{6}\right) + 4\left(\frac{4}{3}\right) - 2\left(\frac{11}{6}\right)^2 - \frac{4^2}{3} \\ &= \frac{91}{6} \end{aligned}$$

Q. 2

$$\text{max. } Z = 2x_1^2 - 7x_2^2 + 12x_1x_2$$

$$\text{sub. to: } 2x_1 + 5x_2 \leq 98 \quad x_1, x_2 \geq 0$$

$$f(x_1, x_2) = 2x_1^2 - 7x_2^2 + 12x_1x_2$$

$$h(x_1, x_2) = 2x_1 + 5x_2 - 98$$

$$\frac{\partial f}{\partial x_1} - \lambda \frac{\partial h}{\partial x_1} = 0$$

$$4x_1 + 12x_2 - 2\lambda = 0 \quad \text{--- 1}$$

$$\frac{\partial f}{\partial x_2} - \lambda \frac{\partial h}{\partial x_2} = 0$$

$$-14x_2 + 12x_1 - 5\lambda = 0 \quad \text{--- 2}$$

$$\lambda (2x_1 + 5x_2 - 98) = 0 \quad \text{--- 3}$$

$$2x_1 + 5x_2 - 98 \neq 0 \quad \text{--- 4}$$

$$x_1, x_2, \lambda \geq 0 \quad \text{--- 5}$$

Case 1:  $\lambda = 0$

$$4x_1 + 12x_2 = 0$$

$$-14x_2 + 12x_1 = 0 \quad \neq 0$$

$$12x_1 - 14x_2 = 0$$

Case II:  $\lambda \neq 0$ ,  $2x_1 + 5x_2 = 98$  — (4)

$$4x_1 + 12x_2 - 2\lambda = 0 \quad \div 2$$

$$12x_1 - 14x_2 + 5\lambda = 0$$

$$2x_1 + 6x_2 = \lambda$$

$$12x_1 - 14x_2 + 5(2x_1 + 6x_2) = 0$$

$$12x_1 - 14x_2 + 10x_1 + 30x_2 = 0$$

$$22x_1 - 4x_2 = 0$$

$$2x_1 + 5x_2 = 98$$

$$-49x_2 = -98$$

$$x_2 = 2$$

$$2x_1 - 44(2) = 0$$

$$2x_1 - 88 = 0$$

$$x_1 = 44$$

$$2(44) + 6(2) = \lambda$$

$$88 + 12 = \lambda$$

$$\lambda = 100$$

$$Z = 2x_1^2 - 7x_2^2 + 12(x_1x_2)$$

$$= 2(44)^2 - 7(2)^2 + 12(44 \times 2)$$

$$= 4900$$

Q.3.  $Z = 10x_1 + 10x_2 - x_1^2 - x_2^2$

sub.  $x_1 + x_2 \leq 8$ ,  $-x_1 + x_2 \leq 5$ ,  $x_1, x_2 > 0$

sol<sup>n</sup>:  $f(x) = 10x_1 + 10x_2 - x_1^2 - x_2^2$

$$h_1 = x_1 + x_2 - 8$$

$$h_2 = -x_1 + x_2 - 5$$



$$\frac{\partial f}{\partial x_1} - \lambda_1 \frac{\partial h_1}{\partial x_1} - \lambda_2 \frac{\partial h_2}{\partial x_1} = 0$$

$$10 - 2x_1 - \lambda_1 + \lambda_2 = 0 \quad \text{--- 1}$$

$$\frac{\partial f}{\partial x_2} - \lambda_1 \frac{\partial h_1}{\partial x_2} - \lambda_2 \frac{\partial h_2}{\partial x_2} = 0$$

$$10 - 2x_2 - \lambda_1 - \lambda_2 = 0 \quad \text{--- 2}$$

$$\lambda_1 (x_1 + x_2 - 8) = 0 \quad \text{--- 3}$$

$$\lambda_2 (-x_1 + x_2 - 5) = 0 \quad \text{--- 4}$$

$$x_1 + x_2 - 8 \leq 0 \quad \text{--- 5}$$

$$-x_1 + x_2 - 5 \leq 0 \quad \text{--- 6} \quad x_1, x_2 \geq 0 \quad \text{--- 7}$$

$$\lambda_1, \lambda_2 \geq 0 \quad \text{--- 8}$$

Now, depending on the values of  $x_1$  &  $x_2$  following cases:-

Case I:  $\lambda_1, \lambda_2 = 0$

from 1 & 2

$$10 - 2x_1 = 0$$

$$x_1 = 5$$

$$10 - 2x_2 = 0$$

$$x_2 = 5$$

But these values do not satisfy eq<sup>n</sup> 5 so we reject these values

Case II:  $\lambda_1 = 0, \lambda_2 \neq 0$  ( $-x_1 + x_2 = 5$ )

$$10 - 2x_1 - \lambda_1 + \lambda_2 = 0$$

$$10 - 2x_1 + \lambda_2 = 0 \quad \text{---}$$

$$10 - 2x_2 - \lambda_1 - \lambda_2 = 0$$

$$10 - 2x_2 - \lambda_2 = 0$$

$$10 - 2x_1 + \lambda_2 + 10 - 2x_2 - \lambda_2 = 0$$

$$-2x_1 - 2x_2 + 20 = 0$$

$$x_1 + x_2 = 10$$

$$-x_1 + x_2 = 5$$

$$2x_2 = 15$$

$$x_2 = \frac{15}{2}$$

$$x_1 + \frac{15}{2} = 10$$

$$x_1 = \frac{10 - 15}{2}$$

$$x_1 = \frac{5}{2}$$

$$\frac{5}{2} + \frac{15}{2} - 8 \leq 0$$

$$\frac{20}{2} - 8 \leq 0$$

$$10 - 8 \neq 0$$

So, these values do not satisfy eqns hence rejecting these values

Case 3:-  $\lambda_1 \neq 0$  ( $x_1 + x_2 = 8$ )  $\lambda_2 = 0$

$$10 - 2x_1 - \lambda_1 = 0$$

$$10 - 2x_2 - \lambda_2 = 0$$

$$10 - 2x_1 - \lambda_1 - 10 + 2x_2 + \lambda_2 = 0$$

$$-2x_1 + 2x_2 = 0$$

$$x_1 - x_2 = 0$$

$$x_1 + x_2 = 8$$

$$2x_1 = 8$$

$$x_1 = 4$$

$$x_2 = 4$$

$$x_1 + x_2 - 8$$

$$4 + 4 - 8$$

$$= 0 \therefore \leq 0$$

Hence satisfies



$$\begin{aligned}
 z &= 10x_1 + 10x_2 - x_1^2 - x_2^2 \\
 &= 10(4) + 10(4) - 4^2 - 4^2 \\
 &= 40 + 40 - 16 - 16 \\
 z_{\max} &= 48
 \end{aligned}$$

Case IV:-  $\lambda_1 \neq 0, \lambda_2 \neq 0$   
 $(x_1 + x_2 = 8) \quad (-x_1 + x_2 = 5)$

$$10 - 2x_1 - \lambda_1 + \lambda_2 = 0$$

$$10 - 2x_2 - \lambda_1 - \lambda_2 = 0$$

$$x_1 + x_2 = 8$$

$$-x_1 + x_2 = 5$$

$$2x_2 = 13$$

$$x_2 = \frac{13}{2}$$

$$x_1 + \frac{13}{2} = 8$$

$$x_1 = 8 - \frac{13}{2}$$

$$x_1 = \frac{3}{2}$$

FOC  $10 - 2x_1 - \lambda_1 + \lambda_2 = 0$

~~2nd~~  $10 - 2\left(\frac{13}{2}\right) - \lambda_1 + \lambda_2 = 0$

$$10 - 13 - \lambda_1 + \lambda_2 = 0$$

$$-3 - \lambda_1 + \lambda_2 = 0$$

$$\lambda_1 - \lambda_2 = -3 \quad \text{--- (1)}$$

$$10 - 2\left(\frac{13}{2}\right) - \lambda_1 - \lambda_2 = 0$$

$$10 - 13 - \lambda_1 - \lambda_2 = 0$$

$$\lambda_1 + \lambda_2 = -3 \quad \text{--- (2)}$$

$$\lambda_1 - \lambda_2 = -3$$

$$2\lambda_1 = -6$$

$$\lambda_1 = -3$$

$$2 - \lambda_2 = 7$$

$$\lambda_2 = 2 - 7$$

$$\lambda_2 = -5$$

$\lambda_1 \in$  we reject this values

$$\therefore Z_{\max} = 48$$

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Q. Using Kuhn-Tucker cond<sup>n</sup> solve the given NLP  
 max.  $Z = -2x_1^2 - 2x_2^2 + 12x_1 + 21x_2 + 2x_1x_2$   
 sub.  $x_1 + x_2 \leq 10, x_2 \leq 8, x_1, x_2 \geq 0$

$$\rightarrow f(x) = -2x_1^2 - 2x_2^2 + 12x_1 + 21x_2 + 2x_1x_2$$

$$h_1 = x_1 + x_2 - 10$$

$$h_2 = x_2 - 8$$

$$\frac{\partial f}{\partial x_1} - \lambda_1 \frac{\partial h_1}{\partial x_1} - \lambda_2 \frac{\partial h_2}{\partial x_1} = 0$$

$$-4x_1 + 12 + 2x_2 - \lambda_1 = 0 \quad \text{--- 1}$$

$$\frac{\partial f}{\partial x_2} - \lambda_1 \frac{\partial h_1}{\partial x_2} - \lambda_2 \frac{\partial h_2}{\partial x_2} = 0$$

$$-4x_2 + 21 + 2x_1 - \lambda_1 - \lambda_2 = 0 \quad \text{--- 2}$$

$$\lambda_1 (x_1 + x_2 - 10) = 0 \quad \text{--- 3}$$

$$\lambda_2 (x_2 - 8) = 0 \quad \text{--- 4}$$

$$x_1 + x_2 - 10 \leq 0 \quad \text{--- 5}$$

$$x_2 - 8 \leq 0 \quad \text{--- 6}$$

$$x_1, x_2, \lambda_1, \lambda_2 \geq 0$$

Now, depending on the values following are the cases:-



case 1:-  $\lambda_1, \lambda_2 = 0$

from 1 and 2

$$-4x_1 + 12 + 2x_2 = 0$$

$$-4x_2 + 21 + 2x_1 = 0$$

$$-4x_1 + 2x_2 = -12$$

$$2x_1 - 4x_2 = -21$$

$$-8x_1 + 4x_2 = -24$$

$$2x_1 - 4x_2 = -21$$

$$-6x_1 = -45$$

$$x_1 = \frac{-45}{-6} = +\frac{15}{2}$$

~~$x_2 = 0$~~

$$-4x_1 + 2x_2 = -12$$

$$-4\left(\frac{+15}{2}\right) + 2x_2 = -12$$

$$-30 + 2x_2 = -12$$

$$2x_2 = -42 \quad 18$$

$$x_2 = -21 \quad 9$$

Since these values does not satisfy the eqn 5.

case 2:-  $\lambda_1 = 0, \lambda_2 \neq 0$

from 1 and 2

$$-4x_1 + 12 + 2x_2 = 0$$

$$-4x_2 + 21 + 2x_1 - \lambda_2 = 0$$

$$-4x_1 + 12 + 2x_2 + -4x_2 + 21 + 2x_1 - \lambda_2 = 0$$

$$-4(x_1 + x_2) + 2(x_1 + x_2) + 33 - \lambda_2 = 0$$

$$-4(10) + 2(10) + 33 - \lambda_2 = 0$$

$$-40 + 20 + 33 - \lambda_2 = 0$$

$$20 + 33 - \lambda_2 = 0$$

$$\lambda_2 = 53$$

Since  $x_2 = 8$

$$-4x_1 + 12 + 2x_2 = 0$$

$$-4x_1 + 12 + 2(8) = 0$$

$$-4x_1 + 12 + 16 = 0$$

$$-4x_1 = -28$$

$$x_1 = 7$$

These values does not satisfy the eq<sup>n</sup> 5

case 3:-  $\lambda_1 \neq 0$ ,  $\lambda_2 = 0$ ,  $x_1 + x_2 = 10$   
from 1 and 2

$$-4x_1 + 12 + 2x_2 - \lambda_1 = 0$$

$$-4x_2 + 21 + 2x_1 - \lambda_1 = 0$$

$$-4x_1 + 12 + 2x_2 - \lambda_1 + 4x_2 + 21 + 2x_1 + \lambda_1 = 0$$

$$-4(x_1 + x_2) + 33 + 2(x_1 + x_2) - 2\lambda_1 = 0$$

$$-6x_1 + 6x_2 - 9 = 0$$

$$-6x_1 + 6x_2 = 9$$

$$6x_1 + 6x_2 = 60$$

$$12x_2 = 69$$

$$x_2 = \frac{69}{12} = \frac{23}{4}$$

$$x_1 + x_2 = 10$$

$$x_1 + \frac{23}{4} = 10 \quad x_1 = 10 - \frac{23}{4}$$

$$= \frac{17}{4}$$

$$\frac{17}{4} + \frac{23}{4} \leq 0 \quad \checkmark$$

$$\frac{23}{4} \leq 8 \quad \checkmark$$



$$-4\left(\frac{17}{n}\right) + 12 + 2\left(\frac{23}{n}\right) - \lambda_1 = 0$$

$$-17 + 12 + \frac{23}{2} - \lambda_1 = 0$$

$$\lambda_1 = -5 + \frac{23}{2}$$

$$\lambda_1 = \frac{-10 + 23}{2} = \frac{13}{2}$$

$$-4\left(\frac{23}{n}\right) + 21 + 2\left(\frac{17}{n}\right) - \frac{13}{2} - \lambda_2 = 0$$

$$-23 + 21 + \frac{17}{2} - \frac{13}{2} = \lambda_2$$

$$\lambda_2 = -2 + \frac{4}{2}$$

$$\lambda_2 = 0$$

$$z_{\max} = -2\left(\frac{17}{n}\right)^2 - 2\left(\frac{23}{n}\right)^2 + 12\left(\frac{17}{n}\right) + 21\left(\frac{23}{n}\right)$$

$$+ 2\left(\frac{17}{n}\right)\left(\frac{23}{n}\right)$$

$$= 118.375$$

Case 4:-  $\lambda_1, \lambda_2 \neq 0$   $n_2 = 8$ ,  $n_1 + n_2 = 10$

$$-4n_1 + 21 + 2n_2 - \lambda_1 = 0$$

$$-4n_2 + 21 + 2n_2 - \lambda_1 = 0 \Rightarrow \lambda_2 = 0$$

$$n_1 + n_2 = 10$$

$$n_2 = 8$$

$$n_1 + 8 = 10$$

$$n_1 = 2$$

$$-4(2) + 12 + 2(8) - \lambda_1 = 0$$

$$-8 + 12 + 16 - \lambda_1 = 0$$

$$\lambda_1 = 20$$

$$-4(8) + 21 + 2(2) - \lambda_1 - \lambda_2 = 0$$

$$-32 - 20 + 21 + 4 - 20 - \lambda_2 = 0$$

$$\lambda_2 = -27$$

These values does not satisfy eqn 8  
hence we reject

### \* Applications of lagrange multiplier

- Lagrange multiplier are a game changer in optimization problem across various fields, they help us to find the best soln while juggling multiple constraints making them super useful in real-world scenarios where resources are limited from maximizing profits to designing efficient machines, this tools help us make smart decision in complex situations
- Economic Applications:-
  - Lagrange multipliers used to optimize resource allocation in economics, helps determine most efficient distribution of limited resources (labour, capital, raw-materials) to maximize output or minimize cost.
  - Production optimization involves finding production levels to maximize profit or minimize cost. Constraints include available resources, production



capacities and market demand

Eg:- A company wants to maximize profit by determining optimal production quantities for 2 products, given limited resources, raw-material and production time

- Utility maximization focuses on maximizing consumer satisfaction or utility
  - consumers aim to allocate their budget to achieve the highest level of satisfaction.
  - constraint include budget limitations and prices of goods

Eg A consumer wants to maximize their utility by allocating a fixed budget between 2 goods, considering their price and preferences

#### \* Application of Kuhn-Tucker

- The Kuhn-Tucker conditions are a set of necessary conditions for optimality in constraint optimization problem, particularly those involving inequality constraints, they are an extension of the Lagrange's multiplier method to handle inequality constraints and are widely used in various fields like Economics, Finance, Engineering & computer science.

#### \* Computer Science

- Machine learning :- optimizing model parameters in ML algorithm, specially those with regularization term or constraint on model complexity.
- Support vector Machine (SVM) :- KKT conditions



are crucial for understanding the dual formulation of SVMs and identifying support vectors

- Engineering:-

- Design optimization:- Designing products optimization or systems (Eg:- structures, circuits) + make performance requirement while minimizing cost or weight subject to physical limitations

- Economics:-

- Consumer theory:- Determining optimal consumption bundles (maximizing utility) subject to budget constraints

- Production planning:- Optimizing production level to minimize cost or maximize profits, considering capacity constraints

- Resource allocation:- Analyzing how to allocate limited resources efficiently across different activities or sectors

Q.1. Explain Lagrange's multiplier method with suitable example

Q.2. Explain KKT with a suitable example



## Unit - 2

### Linear Programming Problem

- LPP Linear programming or linear optimization method that help us to find the optimum solution of the problem. An optimum sol<sup>n</sup> is the solution that is the best possible outcome of a given particular problem.
- In simpler terms, it is the method to find out how to do something in the best possible way with limited resources, you need to do the optimum utilization of resources and achieve <sup>the</sup> best possible result in a particular objective, such as the least cost, the highest margin or the least time.
- The situation that require a search for the best values of the variables subject to certain constraints is where we use linear programming problems. This situation cannot be handled by the usual calculus and numerical techniques.

\* Linear programming Simplex method  
One of the most common method to solve the LPP is the simplex method. In this method we repeat a specific condition 'n' several times until an optimal solution is achieved. The steps required to solve LPP using simplex method are:-

Step 1:- Formulate the LPP based on the given constraints.

Step 2:- Convert all the given inequalities to equations or equalities of the LPP by adding the slack variable to each inequality wherever required.



step 3: Construct the initial simplex table by representing each constraint equation in a row and writing the objective function at the bottom row. The table so obtained is called the simplex table obtained.

step 4:- Identify the greatest negative entry in the bottom row, the column of the element with the highest negative entry is called the pivot column.

step 5:- Divide the entries of the right most column with the entries of the respective pivot column, excluding the entries of the bottommost row. Now the row containing the least entry is the called the pivot row. The pivot element is obtained by the intersection of the pivot row and the pivot column.

step 6:- Using the matrix operation and with <sup>the</sup> help of the pivot element make all the entries in the pivot column to be 0.

step 7:- Check for the non-negative entries in the bottommost row. If there are no negative entries in the bottom row, end the process else start the process again from step 4.

step 8:- The final simplex table so obtained gives the solution to our problem.



Q.1.

$$\text{MAX. } Z = x_1 + 2x_2 + 3x_3$$

$$\text{Sub. } 7x_1 + x_2^{+1.5} \leq 6$$

$$x_1 + 2x_2 \leq 20$$

$$3x_2 + 4x_3 \leq 30$$

Standard form

$$Z - x_1 - 2x_2 - 3x_3 + 0s_1 + 0s_2 + 0s_3 = 0$$

$$7x_1 + x_2 + 0s_1 + 0s_2 + 0s_3 = 6$$

$$x_1 + 2x_2 + 0s_1 + s_2 + 0s_3 = 20$$

$$3x_2 + 4x_3 + 0s_1 + 0s_2 + s_3 = 30$$

$$C_j \quad 1 \quad 2 \quad 3 \quad 0 \quad 0 \quad 0$$

$$C_B \quad B.V. \quad x_1 \quad x_2 \quad x_3 \quad s_1 \quad s_2 \quad s_3 \quad \text{Ratio}$$

$$0 \quad s_1 \quad \boxed{7} \quad 1 \quad \boxed{0} \quad \textcircled{1} \quad 0 \quad 0 \quad 6 \quad C/B = 0.14$$

$$0 \quad s_2 \quad 1 \quad 2 \quad 0 \quad 0 \quad 1 \quad 0 \quad 20 \quad 20/0 = 20$$

$$0 \quad s_3 \quad \textcircled{3} \quad 3 \quad \boxed{4} \quad 0 \quad 0 \quad 1 \quad 30 \quad 30/4 = 7.5$$

$$Z_j \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0$$

$$C_j - Z_j \quad 1 \quad 2 \quad 3 \quad 0 \quad 0 \quad 0$$

Iteration 1:-  $s_1$  leaves  $x_3$  enters

|       |                             |       |       |       |       |       |       |         |       |
|-------|-----------------------------|-------|-------|-------|-------|-------|-------|---------|-------|
| $C_j$ |                             | 1     | 2     | 3     | 0     | 0     | 0     |         |       |
| $C_B$ | $B_V$                       | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $sol^n$ | Ratio |
| 3     | <del><math>x_B</math></del> | 7     | 1     | 1     | 1     | 0     | 0     | 6       |       |
| 0     | $s_2$                       | 1     | 2     | 0     | 0     | 1     | 0     | 20      |       |
| 0     | $s_3$                       | -28   | -1    | 0     | -4    | 0     | 1     | 6       |       |

$[s_1]$

New value = Old value -  $\frac{\text{key row} \times \text{key col.}}{\text{key ele.}}$

for  $s_1$  divide all value divide by key ele

for  $[s_2]$

$$x_1 = 1 - \frac{(1) \times (0)}{1} \quad x_2 = 2 - \frac{(1) \times (0)}{1}$$

$$= 1 \quad = 2$$

$$x_3 = 0 - \frac{(1) \times (0)}{1} \quad s_1 = 0 - \frac{(1) \times (0)}{1}$$

$$= 0 \quad = 0$$

$$s_2 = 1 - \frac{(1) \times (0)}{1} \quad s_3 = 0 - \frac{(1) \times (0)}{1}$$

$$= 1 \quad = 0$$

$$sol^n = 20 - \frac{(1) \times (0)}{1}$$

$$= 20$$

$$Z_j = C_B + \sum (C_B \times x_j)$$

|           |     |    |   |    |   |   |
|-----------|-----|----|---|----|---|---|
| $Z_j$     | 21  | 3  | 3 | 3  | 0 | 0 |
| $G - Z_j$ | -20 | -1 | 0 | -3 | 0 | 0 |



for  $s_3$

$$\begin{aligned} \eta_1 &= \frac{0 - (7) \times (4)}{1} & \eta_2 &= \frac{3 - (17) \times (4)}{1} \\ &= -28 & &= -1 \end{aligned}$$

$$\begin{aligned} \eta_3 &= \frac{4 - (17) \times (4)}{1} & s_1 &= \frac{0 - (17) \times (4)}{1} \\ &= 0 & &= -4 \end{aligned}$$

$$\begin{aligned} s_2 &= \frac{0 - (0) \times (4)}{1} & s_3 &= \frac{1 - (0) \times (4)}{1} \\ &= 0 & &= 1 \end{aligned}$$

$$\begin{aligned} \text{sol}^n &= \frac{30 - (6) \times (4)}{1} \\ &= 6 \end{aligned}$$

hence the optimal sol<sup>n</sup> is obtained

$$\eta_1 = 0, \eta_2 = 0, \eta_3 = 6$$

$$Z = \eta_1 + 2\eta_2 + 3\eta_3$$

$$\boxed{Z = 18}$$

Q2. Max.  $Z = 12\eta_1 + 16\eta_2$

Sub.  $10\eta_1 + 20\eta_2 \leq 120, 8\eta_1 + 8\eta_2 \leq 80$   
 $\eta_1, \eta_2 \geq 0$

|   |             |          |          |       |       |                |
|---|-------------|----------|----------|-------|-------|----------------|
|   | $C_j$       | 12       | 16       | 0     | 0     |                |
|   | $BV$        | $\eta_1$ | $\eta_2$ | $s_1$ | $s_2$ | $\text{Sol}^n$ |
| 0 | $s_1$       | 10       | 20       | 1     | 0     | 120            |
| 0 | $s_2$       | 8        | 8        | 0     | 1     | 80             |
|   | $Z_j$       | 0        | 0        | 0     | 0     |                |
|   | $C_j - Z_j$ | 12       | 16       | 0     | 0     |                |

Ratio

$$120/20 = 6$$

$$80/8 = 10$$

Iteration 1:-  $S_1$  leaves,  $x_2$  enters

|             |       |       |       |       |      |              |
|-------------|-------|-------|-------|-------|------|--------------|
| $C_j$       | 12    | 16    | 0     | 0     |      |              |
| $B_V$       | $x_1$ | $x_2$ | $S_1$ | $S_2$ | Soln | Ratio        |
| 16 $x_2$    | 1/2   | 1     | 1/20  | 0     | 6    | 6 / 1/2 = 12 |
| 0 $S_2$     | 4     | 0     | -2/5  | 1     | 32   | 32 / 4 = 8   |
| $Z_j$       | 8     | 16    | 4/5   | 0     |      |              |
| $C_j - Z_j$ | 4     | 0     | -4/5  | 0     |      |              |

for  $S_2$

$$x_1 = 8 - \frac{(10) \times (8)}{20} \quad x_2 = 8 - \frac{(8) \times (20)}{20}$$

$$= 4 \quad = 0$$

$$S_1 = 0 - \frac{(1) \times (8)}{-20} \quad S_2 = 1 - \frac{(0) \times (8)}{20}$$

$$= \frac{2}{5} \quad = 1$$

$$\text{Soln} = 80 - \frac{(120) \times (8)}{20}$$

$$= 32$$

Iteration 2:-  $S_2$  leaves,  $x_1$  enters

|             |       |       |       |       |      |       |
|-------------|-------|-------|-------|-------|------|-------|
| $C_j$       | 12    | 16    | 0     | 0     |      |       |
| $B_V$       | $x_1$ | $x_2$ | $S_1$ | $S_2$ | Soln | Ratio |
| 16 $x_2$    | 0     | 1     | 1/10  | -1/8  | 2    |       |
| 12 $x_1$    | 1     | 0     | -1/10 | 1/4   | 8    |       |
| $Z_j$       | 12    | 16    | 2/5   | 1     |      |       |
| $C_j - Z_j$ | 0     | 0     | -2/5  | -1    |      |       |



for  $[s_1]$

$$\eta_1 = \frac{1}{2} - \frac{(4) \times (1/2)}{4} \quad \eta_2 = 1 - \frac{(0) \times (1/2)}{4}$$

$$= 0 \quad = 1$$

$$s_1 = \frac{1}{20} - \frac{(-2)}{5} \times \left(\frac{1}{2}\right) \quad s_2 = 0 - (1) \times \left(\frac{1}{2}\right)$$

$$= \frac{1}{20} + \frac{1}{20} \quad = -\frac{1}{8}$$

$$= \frac{2}{20} = \frac{1}{10}$$

Soln:-

$$6 - \frac{(1) \times (1/2)}{4} \quad 6 - \frac{(32) \times (1/2)}{4}$$

$$= 6 - \frac{1}{8} \quad = 6 - 4$$

$$= \frac{47}{8} \quad = 2$$

hence the optimal soln is obtained

$$\eta_1 = 8, \eta_2 = 2$$

$$z_{\max} = 12(12) + 16(2)$$

$$z_{\max} = 400 = 128$$

Q.3. solve the following LPP by simplex method

max.  $z = 6\eta_1 - 2\eta_2 + 3\eta_3$

Sub.  $2\eta_1 - \eta_2 + 2\eta_3 \leq 2$   $\eta_1, \eta_2, \eta_3 \geq 0$

$$\eta_1 + 4\eta_3 \leq 4$$

Standard form

$$z - 6x_1 + 2x_2 - 5x_3 + 0s_1 + 0s_2 + 0s_3 = 0$$

$$2x_1 - x_2 + 2x_3 + s_1 + 0s_2 = 2$$

$$x_1 + 4x_3 + 0s_1 + s_2 = 4$$

|   | $C_j$       | 6     | -2    | 3     | 0     | 0     | sol <sup>n</sup> | Ratio     |
|---|-------------|-------|-------|-------|-------|-------|------------------|-----------|
|   | $B_j$       | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ |                  |           |
| 0 | $s_1$       | 2     | -1    | 2     | 1     | 0     | 2                | $2/2 = 1$ |
| 0 | $s_2$       | 1     | 0     | 4     | 0     | 1     | 4                | $4/1 = 4$ |
|   | $Z_j$       | 0     | 0     | 0     | 0     | 0     |                  |           |
|   | $C_j - Z_j$ | 6     | -2    | 3     | 0     | 0     |                  |           |

$s_1$  leaves,  $x_1$  enters

| $s_1$<br>key element | $C_j$ | 6     | -2    | 3     | 0     | 0     | sol <sup>n</sup> | Ratio       |
|----------------------|-------|-------|-------|-------|-------|-------|------------------|-------------|
|                      | $B_j$ | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ |                  |             |
| 6                    | $x_1$ | 1     | -1/2  | 1     | 1/2   | 0     | 1                | $1/1/2 = 2$ |
| 0                    | $s_2$ | 0     | 1/2   | 3     | -1/2  | 1     | 3                | $3/1/2 = 6$ |

for  $[s_2]$

$$x_1 = 1 - \frac{(2)(1)}{2} \quad x_2 = 0 - \frac{(-1) \times (1)}{2}$$

$$= 0 \quad = \frac{1}{2}$$

$$x_3 = 4 - \frac{(2)(1)}{2} \quad s_1 = 0 - \frac{(1) \times (1)}{2}$$

$$= 3 \quad = -\frac{1}{2}$$

$$s_2 = 1 - \frac{(0) \times (1)}{2} = 1$$



$$\text{Soln} = \frac{4 - (2)(1)}{2}$$

$$= 3$$

|             |   |                |    |    |   |
|-------------|---|----------------|----|----|---|
| $Z_j$       | 6 | -3             | 6  | 3  | 0 |
| $C_j - Z_j$ | 0 | <del>3</del> 1 | -3 | -3 | 0 |

If  $m_1$  is already enters we check on next min. & value that is 6

$\therefore S_2$  leaves,  $m_2$  enters

~~$S_2$~~   
K.E

|         |             |       |       |       |       |       |      |       |
|---------|-------------|-------|-------|-------|-------|-------|------|-------|
|         | $C_j$       | 6     | -2    | 3     | 0     | 0     |      |       |
|         | $B_j$       | $m_1$ | $m_2$ | $A_B$ | $S_1$ | $S_2$ | Soln | Ratio |
| 6       | $m_1$       | 1     | 0     | 4     | 0     | 1     | 4    |       |
| -2      | $m_2$       | 0     | 1     | 6     | -1    | 2     | 6    |       |
| $24+12$ | $Z_j$       | 6     | -2    | 12    | 2     | 2     |      |       |
|         | $C_j - Z_j$ | 0     | 0     | -9    | -2    | -2    |      |       |

for  $[m_1]$

$$m_1 = 1 - \frac{(0)(-1/2)}{1/2} = 1$$

$$m_2 = \frac{-1}{2} - \frac{(-1/2)(-1/2)}{1/2} = \frac{-1}{2} + \frac{1}{2} = 0$$

$$m_3 = 1 - \frac{(3)(-1/2)}{1/2} = 1 + 3 = 4$$

$$S_1 = \frac{1}{2} - \frac{(-1/2)(-1/2)}{1/2} = \frac{1}{2} - \frac{1}{2} = 0$$

$$S_2 = 0 - (1) \left[ \frac{-1}{2} \right] \times \left[ \frac{2}{1} \right] = 1$$

$$\text{Soln} = 1 - (3) \left[ \frac{-1}{2} \right] \times \left[ \frac{2}{1} \right] = 4$$

$$3 \left[ \frac{-1}{2} \right] \times \left[ \frac{2}{1} \right]$$

$$\left[ \frac{-1}{2} \right] \left[ \frac{-1}{2} \right] \times \left[ \frac{2}{1} \right]$$

$$x_1 = 4 \quad x_2 = 6 \quad x_3 = 0$$

$$\begin{aligned} Z_{\max} &= 6x_1 - 2x_2 + 3x_3 \\ &= 6(4) - 2(6) + 3(0) \\ &= 24 - 12 \\ &= 12 \end{aligned}$$

Q. Solve the LPP by simplex method.

$$\text{min. } Z = x_1 - 3x_2 + x_3$$

$$\text{Sub. } 3x_1 - x_2 + 2x_3 \leq 7$$

$$2x_2 + 4x_2 \geq -12$$

$$x_1, x_2, x_3 \geq 0$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10$$

standard form

$$Z = x_1 - 3x_2 + x_3 + 0s_1 + 0s_2 + 0s_3$$

$$\text{sub. } 3x_1 - x_2 + 2x_3 + s_1 + 0s_2 + 0s_3 \leq 7$$

$$-2x_2 - 4x_2 \leq 12 \quad (\text{convert to max.})$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10$$

|   | $C_j$       | 1     | -3    | 1     | 0     | 0     | 0     |       |               |
|---|-------------|-------|-------|-------|-------|-------|-------|-------|---------------|
|   | $B_i$       | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | sol'n | Ratio         |
| 0 | $s_1$       | 3     | -1    | 2     | 1     | 0     | 0     | 7     | $7/3 = 2.3$   |
| 0 | $s_2$       | -2    | -4    | 0     | 0     | 1     | 0     | 12    | $12/-4 = -3$  |
| 0 | $s_3$       | -4    | 3     | 8     | 0     | 0     | 1     | 10    | $10/8 = 1.25$ |
|   | $Z_r$       | 0     | 0     | 0     | 0     | 0     | 0     |       |               |
|   | $C_j - Z_j$ | 1     | -3    | 1     | 0     | 0     | 0     |       |               |

$s_1$  leaves,  $x_2$  enters



|             |       |       |       |       |       |       |         |       |
|-------------|-------|-------|-------|-------|-------|-------|---------|-------|
| $C_j$       | 1     | -3    | 1     | 0     | 0     | 0     |         |       |
| $B_v$       | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $sol^n$ | Ratio |
| 0           | $s_1$ | 5/3   | 0     | 14/3  | 1     | 0     | 1/3     | 31/3  |
| 0           | $s_2$ | -22/3 | 0     | 32/3  | 0     | 1     | 4/3     | 76/3  |
| -3          | $x_2$ | -4/3  | 1     | 8/3   | 0     | 0     | 1/3     | 10/3  |
| $z_j$       |       | -4/3  | -8/3  | 0     | 0     | -1/3  |         |       |
| $C_j - z_j$ |       | -3    | 0     | 9     | 0     | 0     | 1       |       |

for  $[s_1]$

$$m_1 = 3 - \frac{(-4)(-1)}{3} \quad m_2 = -1 - \frac{(3)(-1)}{3}$$

$$= 3 - \frac{4}{3}$$

$$= -1 + 1$$

$$= \frac{5}{3}$$

$$= 0$$

$$m_3 = 2 - \frac{(8)(-1)}{3}$$

$$s_1 = 1 - \frac{(0)(-1)}{3}$$

$$= 2 + \frac{8}{3}$$

$$= 1$$

$$= \frac{6+8}{3} = \frac{14}{3}$$

$$s_2 = 0 - \frac{(0)(-1)}{3}$$

$$s_3 = 0 - \frac{(1)(-1)}{3}$$

$$= 0$$

$$= \frac{1}{3}$$

$$sol^n = 7 - \frac{(10)(-1)}{3}$$

$$= 7 + \frac{10}{3}$$

$$= \frac{31}{3}$$

for  $[s_2]$

$$\eta_1 = \frac{(-2) - (-4)(-4)}{3}$$

$$= \frac{-2 - 16}{3}$$

$$= \frac{-6 - 16}{3} = \frac{-22}{3}$$

$$\eta_2 = \frac{-4 - (5)(-4)}{3}$$

$$= \frac{-4 + 20}{3}$$

$$= 0$$

$$\eta_3 = \frac{0 - (8)(-4)}{3}$$

$$= \frac{32}{3}$$

$$s_1 = \frac{0 - (0)(-4)}{3}$$

$$= 0$$

$$s_2 = \frac{1 - (0)(-4)}{3}$$

$$= 1$$

$$s_3 = \frac{0 - (1)(-4)}{3}$$

$$= \frac{4}{3}$$

$$sol^n = \frac{12 - (10)(-4)}{3}$$

$$= \frac{12 + 40}{3}$$

$$= \frac{36 + 40}{3}$$

$$= \frac{76}{3}$$

$s_1$  leaves,  $\eta_1$  enters.



$$\begin{array}{r|rrrrrrrr}
 CJ & 1 & -3 & 1 & 0 & 0 & 0 & \\
 BV & n_1 & n_2 & n_3 & s_1 & s_2 & s_3 & sol^n \quad Ratio \\
 1 & n_1 & 1 & 0 & 14/5 & 3/5 & 0 & 1/5 & 31/5 \\
 0 & s_2 & ~~-22/5~~ & 0 & 156/5 & 22/5 & 1 & 14/5 & 354/5 \\
 -3 & n_2 & 0 & 1 & 32/5 & 4/5 & 0 & 3/5 & 58/5 \\
 Z_j & 1 & -3 & -82/5 & -9/5 & 0 & -8/5 & & \\
 \hline
 \text{for } s_2 & 0 & 0 & 87/5 & 9/5 & 0 & 8/5 & & 
 \end{array}$$

for  $[s_2]$

$$n_1 = \frac{-22}{3} - \underbrace{\left(\frac{8}{3}\right) \times \left(\frac{1}{3}\right)}_{(5/3)} \left[\frac{-22}{3}\right] \quad n_2 = 0 - \underbrace{(0) \times \left[\frac{-22}{3}\right]}_{5/3}$$

$$\begin{aligned}
 &= \frac{-22}{3} + \frac{5}{3} \frac{22}{3} \\
 &= \frac{-22}{3} \quad 0
 \end{aligned}$$

$$n_3 = \frac{32}{3} - \underbrace{\left[\frac{14}{3}\right] \times \left[\frac{-22}{3}\right]}_{(5/3)} \quad s_1 = 0 - \underbrace{(1) \times \left(\frac{-22}{3}\right)}_{5/3}$$

$$= \frac{32}{3} + \frac{(308)}{15}$$

$$= \frac{(160 + 308)}{15}$$

$$= \frac{468}{15} = \frac{156}{5}$$

$$s_2 = 1 - \underbrace{(0) \times \left(\frac{-22}{3}\right)}_{5/3}$$

$$= 1$$

$$s_3 = \frac{4}{3} - \underbrace{\left(\frac{1}{3}\right) \times \left(\frac{-22}{3}\right)}_{5/3}$$

$$= \frac{4}{3} + \frac{22}{15}$$

$$= \frac{20 + 22}{15} = \frac{42}{15} = \frac{14}{5}$$

$$sol^n = \frac{76}{3} - \underbrace{\left[\frac{31}{3}\right] \times \left[\frac{-22}{3}\right]}_{5/3}$$

$$= \frac{76}{3} + \frac{682}{15}$$

$$= \frac{380 + 682}{15}$$

$$= \frac{1062}{15} = \frac{354}{5}$$

for  $[x_2]$

$$x_1 = \frac{-4}{3} - \left( \frac{5}{3} \right) \times \left( \frac{-4}{3} \right) \quad x_2 = \frac{1 - (0) \left( \frac{-4}{3} \right)}{5/3}$$

$$= \frac{-4}{3} + \frac{20}{3} = 0 \quad = 1$$

$$x_3 = \frac{8}{3} - \left( \frac{14}{3} \right) \times \left( \frac{-4}{3} \right) \quad s_1 = \frac{0 - (1) \left( \frac{-4}{3} \right)}{5/3}$$

$$= \frac{8}{3} + \frac{56}{15} \quad = \frac{4}{5}$$

$$= \frac{40 + 56}{15} = \frac{96}{15} = \frac{32}{5}$$

$$s_2 = \frac{0 - (0) \times \left( \frac{-4}{3} \right)}{5/3} \quad s_3 = \frac{1 - \left( \frac{1}{3} \right) \left( \frac{-4}{3} \right)}{5/3}$$

$$= 0 \quad = \frac{1}{3} + \frac{4}{15} = \frac{3}{15} + \frac{4}{15} = \frac{7}{15}$$

$$s_{01} = \frac{10 - \left( \frac{31}{3} \right) \times \left( \frac{-4}{3} \right)}{5/3}$$

$$= \frac{10}{3} + \frac{124}{15} = \frac{50 + 124}{15} = \frac{174}{15} = \frac{58}{5}$$



All  $G-Z_j$  are  $\geq 0$

Hence the optimal sol<sup>n</sup> obtained

$$n_1 = 31/5, n_2 = 58/5, n_3 = 0$$

$$= n_1 - 3n_2 + n_3$$

$$= \frac{31}{5} - 3\left(\frac{58}{5}\right) + 0$$

$$= \frac{-143}{5}$$

Q. max.  $Z = 4n_1 + 2n_2 + 5n_3$

Sub.  $12n_1 + 7n_2 + 9n_3 \leq 1260$

$22n_1 + 18n_2 + 16n_3 \leq 19008$

$2n_1 + 4n_2 + 3n_3 \leq 396$

$n_1, n_2, n_3 \geq 0$

|   | $C_j$   | 4     | 2     | 5     | 0     | 0     | 0     |                  |                 |
|---|---------|-------|-------|-------|-------|-------|-------|------------------|-----------------|
|   | $B_i$   | $n_1$ | $n_2$ | $n_3$ | $s_1$ | $s_2$ | $s_3$ | Sol <sup>n</sup> | Ratio           |
| 0 | $s_1$   | 12    | 7     | 9     | 1     | 0     | 0     | 1260             | $1260/9=140$    |
| 0 | $s_2$   | 22    | 18    | 16    | 0     | 1     | 0     | 19008            | $19008/16=1188$ |
| 0 | $s_3$   | 2     | 4     | 3     | 0     | 0     | 1     | 396              | $396/3=132$     |
|   | $Z_j$   | 0     | 0     | 0     | 0     | 0     | 0     |                  |                 |
|   | $G-Z_j$ | 4     | 2     | 5     | 0     | 0     | 0     |                  |                 |

$s_3$  leaves  $n_3$  enters

|   | $C_j$   | 4      | 2       | 5     | 0     | 0     | 0       |                  |                   |
|---|---------|--------|---------|-------|-------|-------|---------|------------------|-------------------|
|   | $B_i$   | $n_1$  | $n_2$   | $n_3$ | $s_1$ | $s_2$ | $s_3$   | Sol <sup>n</sup> | Ratio             |
| 0 | $s_1$   | 6      | -5      | 0     | 1     | 0     | -3      | 72               | $72/6=12$         |
| 0 | $s_2$   | $74/3$ | $-10/3$ | 0     | 0     | 1     | $-16/3$ | 16896            | $16896/34/3=1494$ |
| 5 | $n_3$   | $2/3$  | $4/3$   | 1     | 0     | 0     | $1/3$   | 132              | $132/1/3=132$     |
|   | $Z_j$   | $10/3$ | $20/3$  | 5     | 0     | 0     | $5/3$   |                  |                   |
|   | $G-Z_j$ | $2/3$  | $-44/3$ | 0     | 0     | 0     | $-5/3$  |                  |                   |

for  $[S_1]$

$$\begin{aligned} \eta_1 &= \frac{12 - (2)(9)}{3} \\ &= \frac{12 - 6}{3} \\ &= 6 \end{aligned}$$

$$\begin{aligned} \eta_2 &= \frac{7 - (4)(9)}{3} \\ &= \frac{7 - 12}{3} \\ &= -5 \end{aligned}$$

$$\begin{aligned} \eta_3 &= \frac{9 - (3)(9)}{3} \\ &= 0 \end{aligned}$$

$$\begin{aligned} s_1 &= \frac{1 - (0)(9)}{3} \\ &= 1 \end{aligned}$$

$$\begin{aligned} s_2 &= \frac{0 - (0)(9)}{3} \\ &= 0 \end{aligned}$$

$$\begin{aligned} s_3 &= \frac{0 - (1)(9)}{3} \\ &= -3 \end{aligned}$$

$$\begin{aligned} \text{Soln} &= \frac{1260 - (396)(9)}{3} \\ &= \frac{1260 - 1188}{3} \\ &= 72 \end{aligned}$$

for  $[S_2]$

$$\begin{aligned} \eta_1 &= \frac{22 - (2)(16)}{3} \\ &= \frac{22 - 32}{3} \\ &= -\frac{10}{3} \end{aligned}$$

$$\begin{aligned} \eta_2 &= \frac{18 - (4)(16)}{3} \\ &= \frac{18 - 64}{3} \\ &= -\frac{46}{3} \end{aligned}$$

$$\begin{aligned} \eta_3 &= \frac{16 - (3)(16)}{3} \\ &= 0 \end{aligned}$$

$$\begin{aligned} s_1 &= \frac{0 - (0)(16)}{3} \\ &= 0 \end{aligned}$$



$$S_2 = 1 - \frac{(0) \times (16)}{3}$$

$$= 1$$

$$S_3 = 0 - \frac{(1)(16)}{3}$$

$$= -\frac{16}{3}$$

$$Sol^n = 19008 - \frac{(396)(16)}{3}$$

$$= 19008 - (132)(16)$$

$$= 16896$$

$S_1$  leaves,  $\eta_1$  enters

|             |          |          |          |          |       |       |       |         |
|-------------|----------|----------|----------|----------|-------|-------|-------|---------|
|             | $C_j$    | 4        | 2        | 5        | 0     | 0     | 0     |         |
|             | $B_j$    | $\eta_1$ | $\eta_2$ | $\eta_3$ | $S_1$ | $S_2$ | $S_3$ | $Sol^n$ |
| 4           | $\eta_1$ | 1        | -5/6     | 0        | 1/6   | 0     | -1/2  | 12      |
| 0           | $S_2$    | 0        | 55/9     | 0        | -17/9 | 1     | 1/3   | 16760   |
| 5           | $\eta_3$ | 0        | 17/9     | 1        | -1/9  | 0     | 2/3   | 124     |
| $z_j$       | 4        | 55/9     | 5        | 11/9     | 0     | 4/3   |       |         |
| $C_j - z_j$ | 0        | -37/9    | 0        | -11/9    | 0     | -4/3  |       |         |

for  $[S_2]$

$$\eta_1 = \frac{34}{3} - \frac{(8)(\frac{34}{3})}{6}$$

$$= 0$$

$$\eta_3 = 0 - \frac{(0) \times (\frac{34}{3})}{6}$$

$$= 0$$

$$\eta_2 = \frac{-10}{3} - \frac{(-5)(\frac{34}{3})}{6}$$

$$= \frac{-10}{3} + \frac{170}{18}$$

$$= \frac{-60 + 170}{18} = \frac{110}{18}$$

$$= \frac{55}{9}$$

$$s_1 = \frac{0 - (1) \left( \frac{34}{3} \right)}{6}$$

$$s_2 = \frac{1 - (0) \times \left( \frac{34}{3} \right)}{6}$$

$$= \frac{-34}{18} = -\frac{17}{9}$$

$$= 1$$

$$s_3 = \frac{-10 - (-8) \left( \frac{34}{3} \right)}{6} \quad \text{soln} = \frac{16896 - 72 \times \left( \frac{34}{3} \right)}{8}$$

$$= \frac{-10 - \frac{-34}{6}}{6} = 16760$$

$$= \frac{-10 + 32 + 34}{6}$$

$$= \frac{2}{6} = \frac{1}{3}$$

for  $[n_3]$

$$n_1 = \frac{2 - (6) \left( \frac{2}{3} \right)}{6}$$

$$n_2 = \frac{9 - (-5) \left( \frac{2}{3} \right)}{6}$$

$$= 0$$

$$= \frac{9 + 10}{3 \cdot 18}$$

$$n_3 = \frac{1 - (0) \left( \frac{2}{3} \right)}{6}$$

$$= \frac{24 + 10}{18} = \frac{34}{18}$$

$$= \frac{17}{9}$$

$$s_1 = \frac{0 - (1) \left( \frac{2}{3} \right)}{6}$$

$$s_2 = \frac{0 - (0) \left( \frac{2}{3} \right)}{6}$$

$$= -\frac{2}{18} = -\frac{1}{9}$$

$$= 0$$



$$s_3 = \frac{1 - (-3)\left(\frac{2}{8}\right)}{6}$$

$$= \frac{1}{3} + \frac{2}{6}$$

$$= \frac{2}{3}$$

$$Sol^n = 132 - (32)\left(\frac{2}{8}\right)$$

$$= 132 - 8$$

$$= 124$$

$$x_1 = 12, \quad x_2 = 0, \quad x_3 = 124$$

$$Z_{max} = 4x_1 + 2x_2 + 5x_3$$

$$= 4(12) + 0 + 5(124)$$

$$= 668$$